

SUBHOJEET CHOWDHURY

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PROFESSIONAL SUMMARY

Software engineer with 2+ years of experience specialising in agentic AI systems, cloud-native AWS architectures, and full-stack enterprise applications. Career arc spans full-stack HR applications → ITSM transformation → multi-agent LangGraph pipelines → autonomous AWS-native WFA systems. Strong foundations in Python, data structures, and system design; growing interest in classical ML, deep learning, and foundation models.

TECHNICAL SKILLS

AI / Agents: Agentic AI · Multi-Agent Systems · LangGraph · LangChain · RAG · Prompt Eng. · HITL Design · LLMs · Guardrails · Azure OpenAI

Cloud — AWS: Bedrock · Lambda · SQS · DynamoDB · S3 · EC2 · IAM

Cloud — GCP: Dialogflow CX · Google ADK · GCP Storage

Languages: Python · C++ · TypeScript · JavaScript · SQL

Frameworks: Angular · React · FastAPI · Flask · REST APIs · JWT

Platforms: ServiceNow · FreshService

Dev & Tools: Docker · Git · CI/CD · Kubernetes · VS Code

CS Foundations: DSA · DBMS · OS · Computer Networks · System Design · OOP

ML & Interests: Classical ML · Probability & Statistics · Linear Algebra · Deep Learning · Speech / TTS Models

EXPERIENCE

Software Engineer — Tata Consultancy Services (TCS)

Jul 2024 – Present

Kolkata, India

Full-Stack & ITSM (Phases 1–2)

- Built end-to-end HR operations suite (Angular/React + Python/Flask) covering recruitment, onboarding, payroll, and off boarding; integrated with ServiceNow for structured ticket routing.
- Configured and extended FreshService for a ~300,000-user blue-collar workforce (Applied Universal), automating ticket classification, routing, and resolution tracking across HR and IT.

Agentic AI (Phase 3)

- Led migration of rule-based HR apps to multi-agent orchestrated workflows using LangGraph and LangChain; built AI chatbot handling employee queries on leave, payroll, and HR policies — cutting help-desk load significantly.
- Deployed RAG pipelines over large HR knowledge bases for grounded, policy-aware responses; built an AI interview-scheduling agent automating slot matching and candidate communication.

Autonomous WFA System (Phase 4 — Current, AWS)

- Designing an end-to-end autonomous Workforce Administration (WFA) system on AWS: Email ingestion → PII masking → Bedrock classification → human review → Oracle HRM updates via SQS, Lambda, S3 and DynamoDB.
- Built document decomposition layer to split multi-request emails into parallel sub-workflows; designed HITL checkpoints for data incompleteness, compliance failures, and approval gates before committing any HRM changes.

KEY PROJECTS [\[https://subhojeet-chowdhury.vercel.app/\]](https://subhojeet-chowdhury.vercel.app/)

SpeakType — Cross-App Voice Dictation Engine

Rust · Whisper.cpp · Python · Llm

Privacy-first, lightning-fast, cross-application speech-to-text tool: OS-level global hotkeys and local Whisper inference run fully on-device, with a cleanup agent applying context-aware tone and formatting before native keystroke injection — zero cloud dependency.

Viva — AI-Powered Interview Examiner

NextJs · FastAPI · LangGraph

Full-stack agentic platform that generates structured interview papers from any topic, conducts voice-driven Q&A, and evaluates spoken answers against LLM-generated rubrics; config-driven multi-provider LLM abstraction with real-time delivery analytics (pace, filler words).

EDUCATION

B.Tech — Computer Science & Engineering

2020 – 2024

Heritage Institute of Technology, Kolkata (MAKAUT) CGPA: 8.71 / 10